NAMAN JAIN

Backend Software Engineer | Distributed Systems | AI Applications

+91 7490994124 | naman.jain.udp04@gmail.com | LinkedIn | Github | Leetcode | Portfolio

SUMMARY

Final-year Computer Science student (CGPA 9.85/10) with experience building production-oriented backend systems, scalable APIs, and distributed applications. Currently working as a Backend Developer Intern at Shipturtle, contributing to SaaS infrastructure for multi-vendor marketplaces. Strong foundations in data structures, operating systems, databases, networking, and system design, with 205+ solved LeetCode problems and experience shipping full-stack products.

EXPERIENCE

Backend Developer Intern
Shipturtle Apps Pvt. Ltd.
July 2026 – Present

- Learned **PHP and Laravel** within one week to quickly contribute to a production SaaS codebase.
- Developing backend features and RESTful APIs for a multi-vendor e-commerce marketplace platform.
- Working with Laravel, MySQL, and third-party integrations to implement scalable business logic.
- Debugging production issues, improving existing modules, and writing clean, maintainable backend code.
- Collaborating with engineers through code reviews, testing, and iterative feature development in an agile environment.

PROJECTS

Snap Attend — AI Attendance System Github Link | Live Link
Tech: Next.js, Node.js, FastAPI, PostgreSQL, Docker

- Designed a **microservice-based architecture** using Node.js and FastAPI, enabling scalable communication between application services and ML inference workloads.
- Developed secure **REST APIs**, database workflows, and image-processing pipelines for reliable real-time attendance tracking and backend performance.
- Containerized services with **Docker** to simplify deployment, improve environment consistency, and support production-ready development workflows.
- Applied **system design principles**, debugging, and API testing to build a maintainable backend capable of handling concurrent requests and future scalability.

Secure RBAC Backend System Github Link | Live Link
Tech: Node.js, Express, MongoDB, Redis

- Engineered a secure backend system implementing **JWT authentication, RBAC authorization, and RESTful APIs** for production-style multi-user applications.
- Improved API performance from approximately **2s to 600ms** through **Redis caching, database optimization, and efficient query design**.
- Strengthened application reliability using **rate limiting, input validation, password hashing, and security-focused middleware** following backend best practices.
- Performed **API testing, debugging, and performance analysis** using Postman and load simulations to ensure stable and maintainable software.

MeetAI — Real-Time AI Collaboration Platform Github Link | Live Link
Tech: Next.js, WebRTC, WebSockets, OpenAI

- Developed a **real-time collaboration platform** using WebSockets and event-driven architecture to support concurrent users with low-latency communication.
- Integrated AI-powered transcription and meeting summarization workflows while designing reliable backend services for asynchronous processing.
- Built authentication, scheduling, and subscription modules using modular backend architecture and scalable API design.
- Explored **distributed system concepts**, debugging real-time synchronization issues and improving reliability across multiple application services.

CLIFlow — C++ Command Line Application Github Link | Live Link

Tech: C++, GitHub Actions

- Developed a high-performance **C++ command-line application** for analyzing shell history using optimized data structures and memory-efficient algorithms.
- Designed modular components with a focus on maintainability, structured logging, and clean software engineering practices.
- Automated build, testing, and release workflows using **GitHub Actions CI/CD**, significantly reducing manual deployment effort.
- Applied algorithmic problem solving and debugging techniques to improve execution efficiency and ensure reliable cross-platform behavior.

Welth — AI Powered Finance Management Github Link | Live Link

Tech: Next.js, Prisma, PostgreSQL

- Built a modular **full-stack SaaS application** with scalable backend architecture supporting authentication, financial data management, and AI-powered insights.
- Designed relational database schemas using **PostgreSQL and Prisma**, emphasizing maintainability, efficient queries, and data consistency.
- Implemented REST APIs, rate limiting, and backend validation to improve application security, reliability, and production readiness.
- Developed interactive dashboards while following clean software architecture principles for future feature scalability.

Federated Learning System Github Link

Tech: Python, PyTorch

- Implemented a **distributed machine learning pipeline** using FedAvg and FedProx to evaluate collaborative model training on decentralized datasets.
- Built reproducible experimentation workflows for non-IID client data while analyzing communication efficiency and model convergence.
- Trained MobileNetV2 on approximately **15,000 medical images**, achieving **98.93% accuracy** through systematic experimentation and evaluation.
- Conducted research on privacy-preserving AI systems, emphasizing scalable architectures, reproducibility, and engineering trade-offs.

Technical Skills

- **Languages:** C++, Java, PHP, Python, JavaScript, TypeScript
- **Backend & Systems:** Node.js, Express.js, Laravel, FastAPI, REST APIs, WebSockets
- **Frontend:** React, Next.js, Tailwind CSS
- **Databases:** MySQL, PostgreSQL, MongoDB, Prisma, Redis
- **DevOps & Cloud:** Docker, AWS, GitHub Actions, CI/CD, Linux, Git
- **AI & ML:** OpenAI API, Gemini API, Federated Learning, Computer Vision
- **CS Fundamentals:** Data Structures & Algorithms, OOP, OS, DBMS, CN, System Design

Education

Marwadi University — B.Tech Computer Science (2023 – 2027) CGPA: **9.85**/10.00

Achievements

- Solved 205+ LeetCode problems and earned the 50-Day LeetCode Badge.
- HackerRank Gold 5-star Java, 1-star C++, and 1-star Problem Solving badges. – HackerRank Link
- Contributed code and fixes to ToolJet's open-source codebase during Hacktoberfest 2024.
- Winner of 2 technical competitions demonstrating engineering problem-solving.
- National and State-Level UCMAS Winner, solving 200 arithmetic problems in 8 minutes.
- Pseudo Core Committee Member, Competitive Programming Club.